where C is the contour bounding the surface S. Thus, Φ can be determined by either Eq. (5.66) or Eq. (5.67), whichever is easier to integrate for the specific problem under consideration.

5-5 Magnetic Properties of Materials

Because of the similarity between the pattern of the magnetic field lines generated by a current loop and those exhibited by a permanent magnet, the loop can be regarded as a magnetic dipole with north and south poles (Section 5-2.2 and **Fig. 5-13**). The magnetic moment $\mathbf{m}$ of a loop of area A has magnitude $m = IA$ and a direction normal to the plane of the loop (in accordance with the right-hand rule). Magnetization in a material is due to atomic scale current loops associated with (1) orbital motions of the electrons and protons around and inside the nucleus and (2) electron spin. The magnetic moment due to proton motion typically is three orders of magnitude smaller than that of the electrons; therefore, the total orbital and spin magnetic moment of an atom is dominated by the sum of the magnetic moments of its electrons.

► The magnetic behavior of a material is governed by the interaction of the magnetic dipole moments of its atoms with an external magnetic field. The nature of the behavior depends on the crystalline structure of the material and is used as a basis for classifying materials as ***diamagnetic***, ***paramagnetic***, or ***ferromagnetic***. ◄

The atoms of a diamagnetic material have no permanent magnetic moments. In contrast, both paramagnetic and ferromagnetic materials have atoms with permanent magnetic dipole moments, albeit with very different organizational structures.

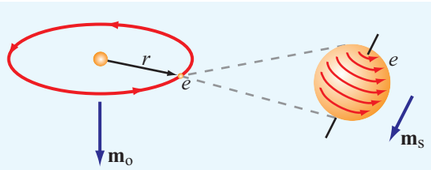
5-5.1 Electron Orbital and Spin Magnetic Moments

This section presents a semiclassical, intuitive model of the atom, which provides quantitative insight into the origin of electron magnetic moments. An electron with charge of $-e$ moving at constant speed u in a circular orbit of radius r (**Fig. 5-20(a)**) completes one revolution in time $T = 2\pi r/u$. This circular motion of the electron constitutes a tiny loop with current I given by

$$I = -\frac{e}{T} = -\frac{eu}{2\pi r}. \tag{5.68}$$

The magnitude of the associated ***orbital magnetic moment*** $\mathbf{m}_o$ is

$$m_o = IA = \left(-\frac{eu}{2\pi r}\right)(\pi r^2) = -\frac{eur}{2} = -\left(\frac{e}{2m_e}\right)L_e, \tag{5.69}$$



(a) Orbiting electron (b) Spinning electron

Figure 5-20 An electron generates (a) an orbital magnetic moment $\mathbf{m}_o$ as it rotates around the nucleus and (b) a spin magnetic moment $\mathbf{m}_s$ as it spins about its own axis.

where $L_e = m_e ur$ is the angular momentum of the electron and m_e is its mass. According to quantum physics, the orbital angular momentum is quantized; specifically, L_e is always some integer multiple of $\hbar = h/2\pi$, where h is Planck's constant. That is, $L_e = 0, \hbar, 2\hbar, \ldots$. Consequently, the smallest nonzero magnitude of the orbital magnetic moment of an electron is

$$m_o = -\frac{e\hbar}{2m_e}. \tag{5.70}$$

Despite the fact that all materials contain electrons that exhibit magnetic dipole moments, most are effectively nonmagnetic. This is because, in the absence of an external magnetic field, the atoms of most materials are oriented *randomly*, as a result of which they exhibit a zero or very small net magnetic moment.

In addition to the magnetic moment due to its orbital motion, an electron has an intrinsic ***spin magnetic moment*** $\mathbf{m}_s$ due to its spinning motion about its own axis (**Fig. 5-20(b)**). The magnitude of $\mathbf{m}_s$ predicted by quantum theory is

$$m_s = -\frac{e\hbar}{2m_e}, \tag{5.71}$$

which is equal to the minimum orbital magnetic moment m_o. The electrons of an atom with an even number of electrons usually exist in pairs with the members of a pair having opposite spin directions, thereby canceling each others' spin magnetic moments. If the number of electrons is odd, the atom has a net nonzero spin magnetic moment due to its unpaired electron.

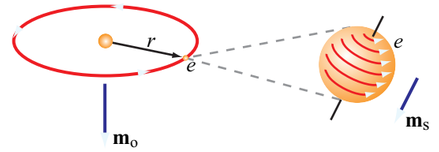
5-5.2 Magnetic Permeability

In Chapter 4, we learned that the relationship $\mathbf{D} = \varepsilon_0\mathbf{E}$, between the electric flux and field in free space, is modified to $\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}$ in a dielectric material. Likewise, the relationship

其中 C 是包围表面 S 的轮廓。因此，Φ 可以由等式 1 确定。（5.66）或等式。（5.67），对于所考虑的特定问题，以更容易积分的方式为准。

5-5 材料的磁性

由于电流环产生的磁力线模式与永磁体产生的磁场线模式相似，因此电流环可以被视为具有北极和南极的磁偶极子（第5-2.2节和图5-13）。面积为 A 的环路的磁矩 m 的大小为 m = IA，方向垂直于环路平面（根据右手定则）。材料中的磁化是由于与 (1) 原子核周围和内部的电子和质子的轨道运动以及 (2) 电子自旋相关的原子级电流环引起的。质子运动产生的磁矩通常比电子的磁矩小三个数量级；因此，原子的总轨道磁矩和自旋磁矩由其电子磁矩之和决定。



(a) Orbiting electron (b) Spinning electron

图 5-20 电子在绕原子核旋转时产生 (a) 轨道磁矩 m_o 和 (b) 在绕自身轴旋转时产生自旋磁矩 m_s。

其中 $L_e = m_e ur$ 是电子的角动量，m_e 是电子的质量。根据量子物理学，轨道角动量是量子化的；具体来说，L_e 始终是 $\hbar = h/2\pi$ 的整数倍，其中 h 是普朗克常数。即，$L_e = 0, \hbar, 2\hbar, \ldots$。因此，电子轨道磁矩的最小非零幅度为

$$m_o = -\frac{e\hbar}{2m_e}. \tag{5.70}$$

► 材料的磁性行为由其原子的磁偶极矩与外部磁场的相互作用控制。行为的性质取决于材料的晶体结构，并用作将材料分类为抗磁性、顺磁性或铁磁性的基础。◄

反磁性材料的原子没有永磁矩。相比之下，顺磁和铁磁材料都具有具有永磁偶极矩的原子，尽管组织结构非常不同。

5-5.1 电子轨道和自旋磁矩

本节介绍了一个半经典、直观的原子模型，它提供了对电子磁矩起源的定量见解。电荷为 −e 的电子在半径为 r 的圆形轨道上以恒定速度 u 运动（图 5-20（a））， 在时间 $T = 2\pi r/u$ 内完成一圈。电子的圆周运动构成了一个微小的环路，电流 I 由下式给出

$$I = -\frac{e}{T} = -\frac{eu}{2\pi r}. \tag{5.68}$$

相关轨道磁矩 m_o 的大小为

$$m_o = IA = \left(-\frac{eu}{2\pi r}\right)(\pi r^2) = -\frac{eur}{2} = -\left(\frac{e}{2m_e}\right)L_e, \tag{5.69}$$

尽管所有材料都含有表现出磁偶极矩的电子，但大多数材料实际上都是非磁性的。这是因为，在没有外部磁场的情况下，大多数材料的原子是随机取向的，因此它们表现出零或非常小的净磁矩。

除了轨道运动产生的磁矩外，电子还因其绕自身轴的自旋运动而具有固有的自旋磁矩m_s（图5-20（b））。量子理论预测的 m_s 的大小为

$$m_s = -\frac{e\hbar}{2m_e}, \tag{5.71}$$

它等于最小轨道磁矩m_o。具有偶数个电子的原子中的电子通常与自旋方向相反的电子对中的成员成对存在，从而抵消彼此的自旋磁矩。如果电子数量为奇数，则原子由于其不成对电子而具有净非零自旋磁矩。

5-5.2 磁导率

在第 4 章中，我们了解到自由空间中电通量与场之间的关系 $D = \varepsilon_0 E$ 在介电材料中被修改为 $D = \varepsilon_0 E + P$。同样，关系

$\mathbf{B} = \mu_0\mathbf{H}$ in free space is modified to

$$\mathbf{B} = \mu_0\mathbf{H} + \mu_0\mathbf{M} = \mu_0(\mathbf{H} + \mathbf{M}), \tag{5.72}$$

where the ***magnetization vector*** $\mathbf{M}$ is defined as the vector sum of the magnetic dipole moments of the atoms contained in a unit volume of the material. Scale factors aside, the roles and interpretations of $\mathbf{B}$, $\mathbf{H}$, and $\mathbf{M}$ in Eq. (5.72) mirror those of $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{P}$ in Eq. (4.93). Moreover, just as in most dielectrics $\mathbf{P}$ and $\mathbf{E}$ are linearly related, in most magnetic materials

$$\mathbf{M} = \chi_m\mathbf{H}, \tag{5.73}$$

where χ_m is a dimensionless quantity called the ***magnetic susceptibility*** of the material. For diamagnetic and paramagnetic materials, χ_m is a (temperature-dependent) constant, resulting in a linear relationship between $\mathbf{M}$ and $\mathbf{H}$ at a given temperature. This is not the case for ferromagnetic substances; the relationship between $\mathbf{M}$ and $\mathbf{H}$ not only is nonlinear, but also depends on the "history" of the material, as explained in the next section.

Keeping this fact in mind, we can combine Eqs. (5.72) and (5.73) to get

$$\mathbf{B} = \mu_0(\mathbf{H} + \chi_m\mathbf{H}) = \mu_0(1 + \chi_m)\mathbf{H}, \tag{5.74}$$

or

$$\mathbf{B} = \mu\mathbf{H}, \tag{5.75}$$

where μ, the ***magnetic permeability*** of the material, relates to χ_m as

$$\mu = \mu_0(1 + \chi_m) \qquad \text{(H/m)}. \tag{5.76}$$

Often it is convenient to define the magnetic properties of a material in terms of the ***relative permeability*** μ_r:

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m. \tag{5.77}$$

A material usually is classified as diamagnetic, paramagnetic, or ferromagnetic on the basis of the value of its χ_m (**Table 5-2**). Diamagnetic materials have negative susceptibilities, whereas paramagnetic materials have positive ones. However, the absolute magnitude of χ_m is on the order of 10^{-5} for both classes of materials, which for most applications allows us to ignore χ_m relative to 1 in Eq. (5.77).

► Thus, $\mu_r \approx 1$ or $\mu \approx \mu_0$ for diamagnetic and paramagnetic substances, which include dielectric materials and most metals. In contrast, $|\mu_r| \gg 1$ for ferromagnetic materials; $|\mu_r|$ of purified iron, for example, is on the order of 2×10^5. ◄

Ferromagnetic materials are discussed next.

Exercise 5-12: The magnetic vector $\mathbf{M}$ is the vector sum of the magnetic moments of all the atoms contained in a unit volume (1m^3). If a certain type of iron with 8.5×10^{28} atoms/m^3 contributes one electron per atom to align its spin magnetic moment along the direction of the applied field, find (a) the spin magnetic moment of a single electron given that $m_e = 9.1 \times 10^{-31}$ (kg) and $\hbar = 1.06 \times 10^{-34}$ (J·s) and (b) the magnitude of $\mathbf{M}$.

Answer: (a) $m_s = 9.3 \times 10^{-24}$ (A·m^2), (b) $M = 7.9 \times 10^5$ (A/m). (See EM.)

5-5.3 Magnetic Hysteresis of Ferromagnetic Materials

Ferromagnetic materials, which include iron, nickel, and cobalt, exhibit unique magnetic properties due to the fact that their magnetic moments tend to readily align along the direction of an external magnetic field. Moreover, such materials remain partially magnetized even after the external field is removed. Because of these peculiar properties, ferromagnetic materials are used in the fabrication of permanent magnets.

A key to understanding the properties of ferromagnetic materials is the notion of ***magnetized domains***, that are microscopic regions (on the order of 10^{-10} m^3) within which the magnetic moments of all atoms (typically on the order of 10^{19} atoms) are permanently aligned with each other. This alignment, which occurs in all ferromagnetic materials, is due to strong coupling forces between the magnetic dipole moments constituting an individual domain. In the absence of an external magnetic field, the domains take on random orientations relative to each other (**Fig. 5-21(a)**), resulting in zero net magnetization. The ***domain walls*** forming the boundaries between adjacent domains consist of thin transition regions. When an unmagnetized sample of a ferromagnetic material is placed in an external magnetic field, the domains partially align with the external field, as illustrated in **Fig. 5-21(b)**. A quantitative understanding of how the domains form and how they behave under the influence of an external magnetic field requires a heavy dose of quantum mechanics and is outside the scope of the present treatment. Hence, we confine our discussion to a qualitative description of the magnetization process and its implications.

The magnetization behavior of a ferromagnetic material is described in terms of its *B–H* ***magnetization curve***, where *B* and *H* refer to the amplitudes of the $\mathbf{B}$ flux and $\mathbf{H}$ field in the material. Suppose that we start with an unmagnetized sample of iron, denoted by point *O* in **Fig. 5-22**. When we increase $\mathbf{H}$ continuously, for example, by increasing the current passing

自由空间中的 B = μ 0H 被修改为

$$\mathbf{B} = \mu_0\mathbf{H} + \mu_0\mathbf{M} = \mu_0(\mathbf{H} + \mathbf{M}), \tag{5.72}$$

其中磁化矢量 M 定义为材料单位体积中所含原子磁偶极矩的矢量和。撇开比例因素不谈，方程中 B、H 和 M 的作用和解释。(5.72) 反映了方程式中的 D、E 和 P。 (4.93) 。此外，正如在大多数电介质中 P 和 E 呈线性相关一样，在大多数磁性材料中

$$\mathbf{M} = \chi_m\mathbf{H}, \tag{5.73}$$

其中 χ_m 是无量纲量，称为材料的磁化率。对于抗磁性和顺磁性材料，χ_m 是一个（与温度相关的）常数，导致给定温度下 M 和 H 之间呈线性关系。铁磁物质则不然。 M 和 H 之间的关系不仅是非线性的，而且还取决于材料的“历史”，如下一节所述。

记住这一事实，我们可以结合等式。 (5.72) 和 (5.73) 得到

$$\mathbf{B} = \mu_0(\mathbf{H} + \chi_m\mathbf{H}) = \mu_0(1 + \chi_m)\mathbf{H}, \tag{5.74}$$

or

$$\mathbf{B} = \mu\mathbf{H}, \tag{5.75}$$

其中 μ （材料的磁导率）与 χ_m 的关系为

$$\mu = \mu_0(1 + \chi_m) \qquad \text{(H/m)}. \tag{5.76}$$

通常，用相对磁导率 μ_r 来定义材料的磁性是很方便的：

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m. \tag{5.77}$$

材料通常根据其χ_m (表 5-2) 的值分为抗磁性、顺磁性或铁磁性。抗磁材料具有负磁化率，而顺磁材料具有正磁化率。然而，对于两类材料，χ_m 的绝对量值约为 10^{-5}，这对于大多数应用来说允许我们忽略等式 1 中相对于 1 的 χ_m。 (5.77) 。

► 因此，$\mu_r \approx 1$ 或 $\mu \approx \mu_0$ 用于抗磁性和顺磁性物质，其中包括介电材料和大多数金属。相反，对于铁磁材料，$|\mu_r| \gg 1$；例如，精铁的 $|\mu_r|$ 约为 2×10^5。 ◄

接下来讨论铁磁材料。

练习5-12：磁矢量M是单位体积（1m^3）中包含的所有原子的磁矩矢量和。如果某种类型的铁具有 8.5×10^{28} 原子/m^3，每个原子贡献一个电子以沿施加场的方向排列其自旋磁矩，则在 $m_e = 9.1 \times 10^{-31}$ (kg) 和 $\hbar = 1.06$ 的情况下，求 (a) 单个电子的自旋磁矩$\times 10^{-34}$ (J·s) 和 (b) M 的大小。

答案：(a) $m_s = 9.3 \times 10^{-24}$ (A·m^2)，(b) $M = 7.9 \times 10^5$ (A/m)。 (参见EM。)

5-5.3 铁磁材料的磁滞现象

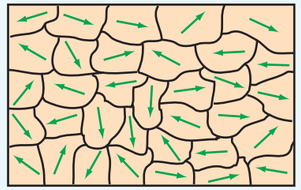
铁磁材料（包括铁、镍和钴）表现出独特的磁性，因为它们的磁矩很容易沿着外部磁场的方向排列。此外，即使在外部磁场移除后，此类材料仍保持部分磁化。由于这些特殊的性质，铁磁材料被用于制造永磁体。

理解铁磁材料特性的关键是磁畴 的概念，磁畴 是微观区域（大约为 10^{-10} m^3），其中所有原子（通常大约为 10^{19} 原子）的磁矩彼此永久对齐。这种排列出现在所有铁磁材料中，是由于构成单个磁域的磁偶极矩之间的强耦合力造成的。在没有外部磁场的情况下，磁畴相对于彼此呈现随机方向（图5-21（a）），导致净磁化强度为零。形成相邻域之间的边界的域壁由薄过渡区域组成。当铁磁材料的未磁化样品放置在外部磁场中时，磁畴部分与外部磁场对齐，如图 5-21(b) 所示。对磁畴如何形成以及它们在外部磁场影响下如何表现的定量理解需要大量的量子力学，并且超出了当前处理的范围。因此，我们的讨论仅限于磁化过程及其含义的定性描述。

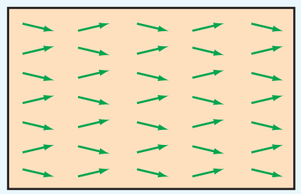
铁磁材料的磁化行为用其 B-H 磁化曲线来描述，其中 B 和 H 是指材料中 B 磁通和 H 场的振幅。假设我们从未磁化的铁样品开始，如图 5-22 中的 O 点所示。当我们不断增加H时，例如，通过增加通过的电流

Table 5-2 Properties of magnetic materials.

	Diamagnetism	Paramagnetism	Ferromagnetism
Permanent magnetic dipole moment	No	Yes, but weak	Yes, and strong
Primary magnetization mechanism	Electron orbital magnetic moment	Electron spin magnetic moment	Magnetized domains
Direction of induced magnetic field (relative to external field)	Opposite	Same	Hysteresis [see Fig. 5-22]
Common substances	Bismuth, copper, diamond, gold, lead, mercury, silver, silicon	Aluminum, calcium, chromium, magnesium, niobium, platinum, tungsten	Iron, nickel, cobalt
Typical value of χ_m **Typical value of μ_r**	$\approx -10^{-5}$ ≈ 1	$\approx 10^{-5}$ ≈ 1	$\lvert\chi_m\rvert \gg 1$ and hysteretic $\lvert\mu_r\rvert \gg 1$ and hysteretic



(a) Unmagnetized domains



(b) Magnetized domains

Figure 5-21 Comparison of (a) unmagnetized and (b) magnetized domains in a ferromagnetic material.

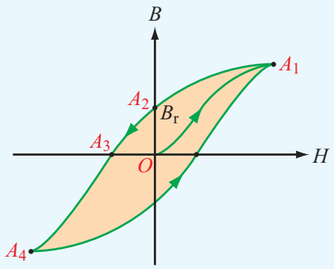
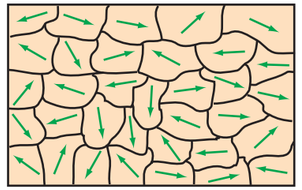


Figure 5-22 Typical hysteresis curve for a ferromagnetic material.

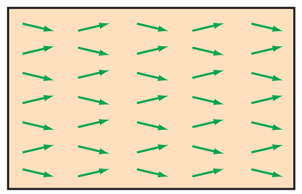
through a wire wound around the sample, **B** increases also along the *B–H* curve from point *O* to point A_1, at which nearly all the domains have become aligned with **H**. Point A_1 represents a saturation condition. If we then decrease **H** from its value at point A_1 back to zero (by reducing the current through the wire), the magnetization curve follows the path from A_1 to A_2. At point A_2, the external field **H** is zero (owing to the fact that the current through the wire is zero), but the flux density **B** in the material is not. The magnitude of **B** at A_2 is called the ***residual flux density*** B_r. The iron material is now magnetized and ready to be used as a permanent magnet owing to the fact that a large fraction of its magnetized domains have remained aligned. Reversing the direction of **H** and increasing its intensity causes **B** to decrease from B_r at point A_2 to zero at point A_3, and if the intensity of **H** is increased further

表5-2 磁性材料的性质

磁性特征	抗磁性	顺磁性	铁磁性
永久磁偶极矩	无	有，较弱	有，较强
主要磁化机制	电子轨道磁矩	电子自旋磁矩	磁畴的磁化
感应磁场相对外场的方向	相反	相同	有磁滞，见图5-22
常见物质	铋、铜、金刚石、金、铅、汞、银、硅	铝、钙、铬、镁、铌、铂、钨	铁、镍、钴
χ_m 的典型值	约 -10^{-5}	约 10^{-5}	$\lvert\chi_m\rvert$ 远大于1，有磁滞
μ_r 的典型值	约1	约1	$\lvert\mu_r\rvert$ 远大于1，有磁滞



(a) 未磁化状态下的磁畴



(b) 磁化状态下的磁畴

图5-21 铁磁材料在 (a) 未磁化状态与 (b) 磁化状态下的磁畴。

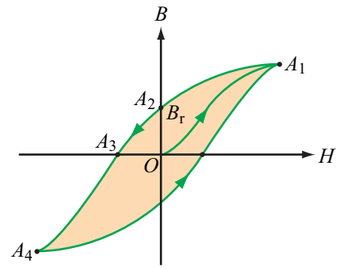
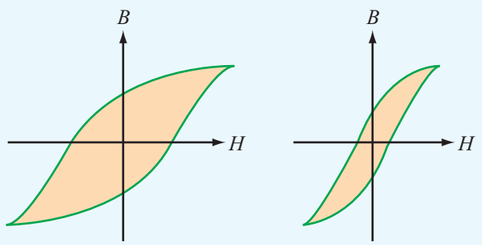


图 5-22 铁磁材料的典型磁滞曲线。

通过缠绕在样品上的导线，B 也沿着 B–H 曲线从 O 点到 A_1 点增加，在该点几乎所有域都与 H 对齐。 A_1 点表示饱和条件。如果我们随后将 H 从 A_1 点的值减小到零（通过减少通过导线的电流），磁化曲线将遵循从 A_1 到 A_2 的路径。在点 A_2 处，外场 H 为零（因为通过导线的电流为零），但材料中的磁通密度 B 并非为零。 A_2 处 B 的大小称为剩余磁通密度 B_r。由于大部分磁畴 保持对齐，铁材料现在已被磁化，可以用作永磁体。反转 H 的方向并增加其强度会导致 B 从 A_2 点处的 B_r 减少到 A_3 点处的零，并且如果 H 的强度进一步增加



(a) Hard material (b) Soft material

Figure 5-23 Comparison of hysteresis curves for (a) a hard ferromagnetic material and (b) a soft ferromagnetic material.

while maintaining its direction, the magnetization moves to the saturation condition at point A_4. Finally, as **H** is made to return to zero and is then increased again in the positive direction, the curve follows the path from A_4 to A_1. This process is called ***magnetic hysteresis***. Hysteresis means "lag behind." The existence of a ***hysteresis loop*** implies that the magnetization process in ferromagnetic materials depends not only on the magnetic field **H** but also on the magnetic history of the material. The shape and extent of the hysteresis loop depend on the properties of the ferromagnetic material and the peak-to-peak range over which **H** is made to vary. ***Hard ferromagnetic materials*** are characterized by wide hysteresis loops (**Fig. 5-23(a)**). They cannot be easily demagnetized by an external magnetic field because they have a large residual magnetization B_r. Hard ferromagnetic materials are used in the fabrication of permanent magnets for motors and generators. ***Soft ferromagnetic materials*** have narrow hysteresis loops (**Fig. 5-23(b)**); hence, they can be more easily magnetized and demagnetized. To demagnetize any ferromagnetic material, the material is subjected to several hysteresis cycles while gradually decreasing the peak-to-peak range of the applied field.

Concept Question 5-12: What are the three types of magnetic materials, and what are typical values of their relative permeabilities?

Concept Question 5-13: What causes magnetic hysteresis in ferromagnetic materials?

Concept Question 5-14: What does a magnetization curve describe? What is the difference between the magnetization curves of hard and soft ferromagnetic materials?

5-6 Magnetic Boundary Conditions

In Chapter 4, we derived a set of boundary conditions that describes how, at the boundary between two dissimilar contiguous media, the electric flux and field **D** and **E** in the first medium relate to those in the second medium. We now derive a similar set of boundary conditions for the magnetic flux and field **B** and **H**. By applying Gauss's law to a pill box that straddles the boundary, we determined that the difference between the normal components of the electric flux densities in two media equals the surface charge density ρ_s. That is,

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q \quad \Rightarrow \quad D_{1n} - D_{2n} = \rho_s. \tag{5.78}$$

By analogy, application of Gauss's law for magnetism, as expressed by Eq. (5.44), leads to the conclusion that

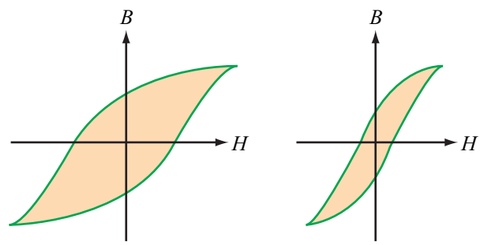
$$\oint_S \mathbf{B}\cdot d\mathbf{s} = 0 \quad \Rightarrow \quad B_{1n} = B_{2n}. \tag{5.79}$$

▶ Thus, the normal component of **B** is continuous across the boundary between two adjacent media. ◀

Because $\mathbf{B}_1 = \mu_1\mathbf{H}_1$ and $\mathbf{B}_2 = \mu_2\mathbf{H}_2$ for linear, isotropic media, the boundary condition for **H** corresponding to Eq. (5.79) is

$$\mu_1 H_{1n} = \mu_2 H_{2n}. \tag{5.80}$$

Comparison of Eqs. (5.78) and (5.79) reveals a striking difference between the behavior of the magnetic and electric fluxes across a boundary: *Whereas the normal component of* **B** *is continuous across the boundary, the normal component of* **D** *is not (unless* $\rho_s = 0$*).* The reverse applies to the tangential components of the electric and magnetic fields **E** and **H**: *Whereas the tangential component of* **E** *is continuous across the boundary, the tangential component of* **H** *is not (unless the surface current density* $\mathbf{J}_s = 0$*).* To obtain the boundary condition for the tangential component of **H**, we follow the same basic procedure used in Chapter 4 to establish the boundary condition for the tangential component of **E**. With reference to **Fig. 5-24**, we apply Ampère's law (Eq. (5.47)) to



(a) Hard material (b) Soft material

图 5-23 (a) 硬铁磁材料和 (b) 软铁磁材料的磁滞曲线比较。

在保持其方向的同时，磁化强度在 A_4 点处达到饱和状态。最后，随着 H 回到零，然后再次向正方向增加，曲线遵循从 A_4 到 A_1 的路径。这个过程称为磁滞。滞后意味着“落后”。磁滞回线的存在意味着铁磁材料的磁化过程不仅取决于磁场 H，还取决于材料的磁历史。磁滞回线的形状和范围取决于铁磁材料的特性以及 H 变化的峰峰值范围。硬铁磁材料的特点是具有宽磁滞回线（图5-23（a））。它们不易被外部磁场消磁，因为它们具有较大的剩余磁化强度 B_r。硬铁磁材料用于制造电动机和发电机的永磁体。软铁磁材料具有窄磁滞回线（图5-23（b））；因此，它们更容易被磁化和消磁。为了对任何铁磁材料进行消磁，该材料会经历几个磁滞循环，同时逐渐减小所施加磁场的峰峰值范围。

概念问题 5-12：磁性材料有哪三种类型？它们的相对磁导率的典型值是多少？

概念问题 5-13：铁磁材料中产生磁滞现象的原因是什么？

概念问题 5-14：磁化曲线描述什么？硬铁磁材料和软铁磁材料的磁化曲线有什么区别？

5-6 磁边界条件

在第四章中，我们导出了一组边界条件，描述了在两种不同的相邻介质之间的边界处，第一种介质中的电通量以及场 D 和 E 与第二种介质中的电通量和场 D 和 E 之间的关系。现在，我们推导出磁通量以及磁场 B 和 H 的一组类似的边界条件。通过将高斯定律应用于跨越边界的药丸盒，我们确定两种介质中电通量密度的法向分量之间的差等于表面电荷密度 ρ_s。那是，

$$\oint_S \mathbf{D}\cdot d\mathbf{s} = Q \quad \Rightarrow \quad D_{1n} - D_{2n} = \rho_s. \tag{5.78}$$

以此类推，应用高斯磁定律，如方程所示。（5.44），得出的结论是

$$\oint_S \mathbf{B}\cdot d\mathbf{s} = 0 \quad \Rightarrow \quad B_{1n} = B_{2n}. \tag{5.79}$$

▶ 因此，B 的法向分量在两个相邻介质之间的边界上是连续的。◀

由于 $B_1 = \mu_1 H_1$ 和 $B_2 = \mu_2 H_2$ 为线性各向同性介质，因此 H 的边界条件对应于等式：(5.79) 是

$$\mu_1 H_{1n} = \mu_2 H_{2n}. \tag{5.80}$$

等式的比较(5.78) 和 (5.79) 揭示了跨越边界的磁通量和电通量行为之间的显着差异：虽然 B 的法向分量在边界上是连续的，但 D 的法向分量却不是（除非 $\rho_s = 0$）。相反的情况适用于电场和磁场 E 和 H 的切向分量：虽然 E 的切向分量在边界上是连续的，但 H 的切向分量却不是（除非表面电流密度 $J_s = 0$）。为了获得 H 切向分量的边界条件，我们遵循第 4 章中建立 E 切向分量边界条件的相同基本程序。参考图 5-24，我们应用 Ampère 定律（方程（5.47））

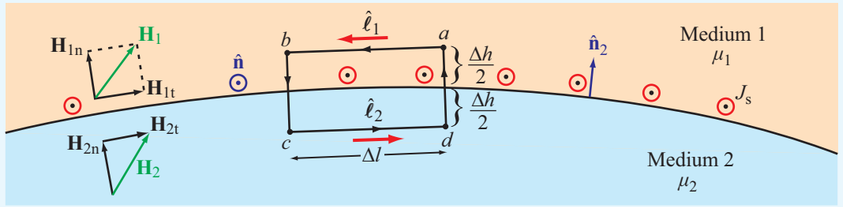


Figure 5-24 Boundary between medium 1 with μ_1 and medium 2 with μ_2.

a closed rectangular path with sides of lengths Δl and Δh, and then let $\Delta h \to 0$ to obtain

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = \int_a^b \mathbf{H}_1\cdot\hat{\boldsymbol{\ell}}_1\, d\ell + \int_c^d \mathbf{H}_2\cdot\hat{\boldsymbol{\ell}}_2\, d\ell = I, \qquad (5.81)$$

where I is the net current crossing the surface of the loop in the direction specified by the right-hand rule (I is in the direction of the thumb when the fingers of the right hand extend in the direction of the loop C). As we let Δh of the loop approach zero, the surface of the loop approaches a thin line of length Δl. The total current flowing through this thin line is $I = J_s\,\Delta l$, where J_s is the magnitude of the component of the surface current density $\mathbf{J}_s$ normal to the loop. That is, $J_s = \mathbf{J}_s\cdot\hat{\mathbf{n}}$, where $\hat{\mathbf{n}}$ is the normal to the loop. In view of these considerations, Eq. (5.81) becomes

$$(\mathbf{H}_1 - \mathbf{H}_2)\cdot\hat{\boldsymbol{\ell}}_1\,\Delta l = \mathbf{J}_s\cdot\hat{\mathbf{n}}\,\Delta l. \qquad (5.82)$$

The vector $\hat{\boldsymbol{\ell}}_1$ can be expressed as $\hat{\boldsymbol{\ell}}_1 = \hat{\mathbf{n}}\times\hat{\mathbf{n}}_2$, where $\hat{\mathbf{n}}$ and $\hat{\mathbf{n}}_2$ are the normals to the loop and to the surface of medium 2 (Fig. 5-24), respectively. Using this relation in Eq. (5.82) and then applying the vector identity $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A})$ leads to

$$\hat{\mathbf{n}}\cdot[\hat{\mathbf{n}}_2\times(\mathbf{H}_1 - \mathbf{H}_2)] = \mathbf{J}_s\cdot\hat{\mathbf{n}}. \qquad (5.83)$$

Since Eq. (5.83) is valid for any $\hat{\mathbf{n}}$, it follows that

$$\hat{\mathbf{n}}_2\times(\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s. \qquad (5.84)$$

This equation implies that the tangential components of $\mathbf{H}$ parallel to $\mathbf{J}_s$ are continuous across the interface, whereas those orthogonal to $\mathbf{J}_s$ are discontinuous in the amount of $\mathbf{J}_s$.

Surface currents can exist only on the surfaces of perfect conductors and superconductors. Hence, *at the interface between media with finite conductivities,* $\mathbf{J}_s = 0$ and

$$H_{1t} = H_{2t}. \qquad (5.85)$$

Example 5-7: Tilted-Plane Boundary

In Fig. 5-25 the plane defined by $y + 2x = 2$ separates medium 1 of permeability μ_1 from medium 2 of permeability μ_2. If no surface current exists on the boundary and

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \qquad \text{(T)},$$

find $\mathbf{B}_2$ and then evaluate the result for $\mu_2 = 2\mu_1$.

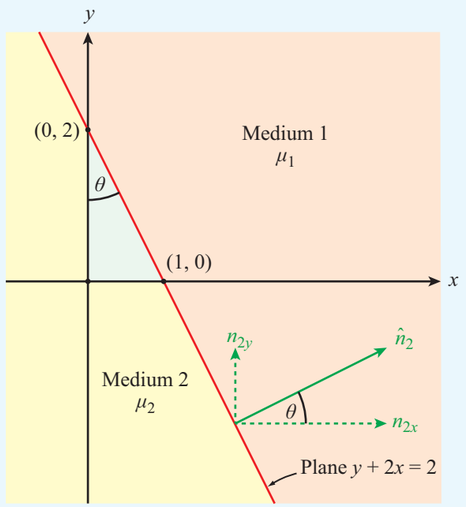


Figure 5-25 Magnetic media separated by the plane $y+2x=2$ (Example 5-7).

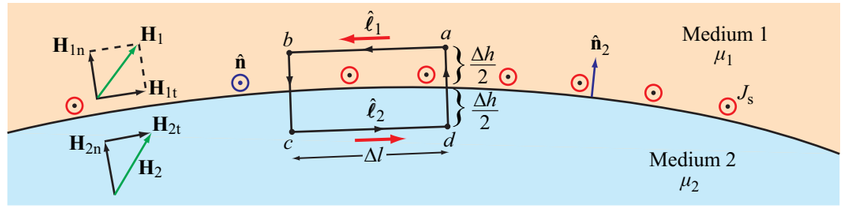


图 5-24 介质 1 与介质 1 之间的边界μ_1和中 2W 它μ_2。

边长为 Δl 和 Δh 的闭合矩形路径，然后令 Δh → 0 得到

$$\oint_C \mathbf{H}\cdot d\mathbf{l} = \int_a^b \mathbf{H}_1\cdot\hat{\boldsymbol{\ell}}_1\, d\ell + \int_c^d \mathbf{H}_2\cdot\hat{\boldsymbol{\ell}}_2\, d\ell = I, \qquad (5.81)$$

其中 I 是按右手定则指定的方向穿过环路表面的净电流（当右手手指沿环路 C 的方向延伸时，I 为拇指方向）。当我们让环路的 Δh 接近零时，环路表面接近一条长度为 Δl 的细线。流过这条细线的总电流为 $I = J_s\,\Delta l$，其中 J_s 是垂直于环路的表面电流密度 J_s 分量的大小。即 $J_s = J_s\cdot\hat{n}$，其中 $\hat{n}$ 是循环的法线。考虑到这些考虑，等式。(5.81) 变为

$$(\mathbf{H}_1 - \mathbf{H}_2)\cdot\hat{\boldsymbol{\ell}}_1\,\Delta l = \mathbf{J}_s\cdot\hat{\mathbf{n}}\,\Delta l. \qquad (5.82)$$

矢量$\hat{\boldsymbol{\ell}}_1$可以表示为$\hat{\boldsymbol{\ell}}_1 = \hat{n}\times\hat{n}_2$，其中$\hat{n}$和$\hat{n}_2$分别是环路和介质2表面的法线（图5-24）。在等式中使用这个关系。(5.82) 然 后应用向量恒等式 $(B\times C) = B\cdot(C\times A)$ 导致

$$\hat{\mathbf{n}}\cdot[\hat{\mathbf{n}}_2\times(\mathbf{H}_1 - \mathbf{H}_2)] = \mathbf{J}_s\cdot\hat{\mathbf{n}}. \qquad (5.83)$$

由于方程。(5.83) 对于任何 $\hat{n}$ 都有效，因此

$$\hat{\mathbf{n}}_2\times(\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s. \qquad (5.84)$$

该方程意味着平行于 J_s 的 H 切向分量在界面上是连续的，而正交于 J_s 的切向分量在 J_s 的量上是不连续的。

表面电流只能存在于理想导体和超导体的表面。因此，在具有有限电导率的介质之间的界面处，$J_s = 0$ 和

$$H_{1t} = H_{2t}. \qquad (5.85)$$

示例 5-7：倾斜平面边界

在图 5-25 中，由 $y + 2x = 2$ 定义的平面将渗透率 μ_1 的介质 1 与渗透率 μ_2 的介质 2 分开。如果边界上不存在表面电流且

$$\mathbf{B}_1 = \hat{\mathbf{x}}2 + \hat{\mathbf{y}}3 \qquad \text{(T)},$$

找到 B_2，然后计算 $\mu_2 = 2\mu_1$ 的结果。

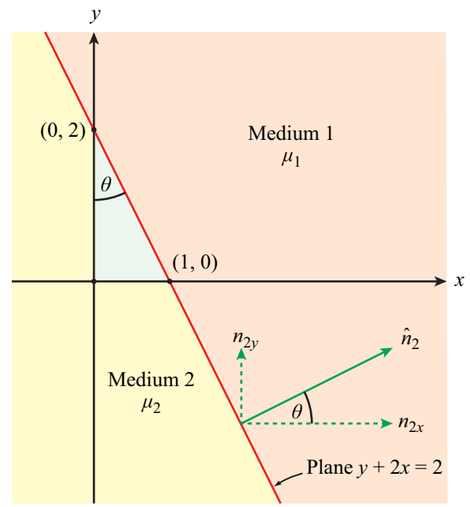


图 5-25 由平面 $y+2x = 2$ 分隔的磁性介质（示例 5-7）。